Modeling Climate Driven Ambystoma tigrinum virus Dynamics

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INTRODUCTION

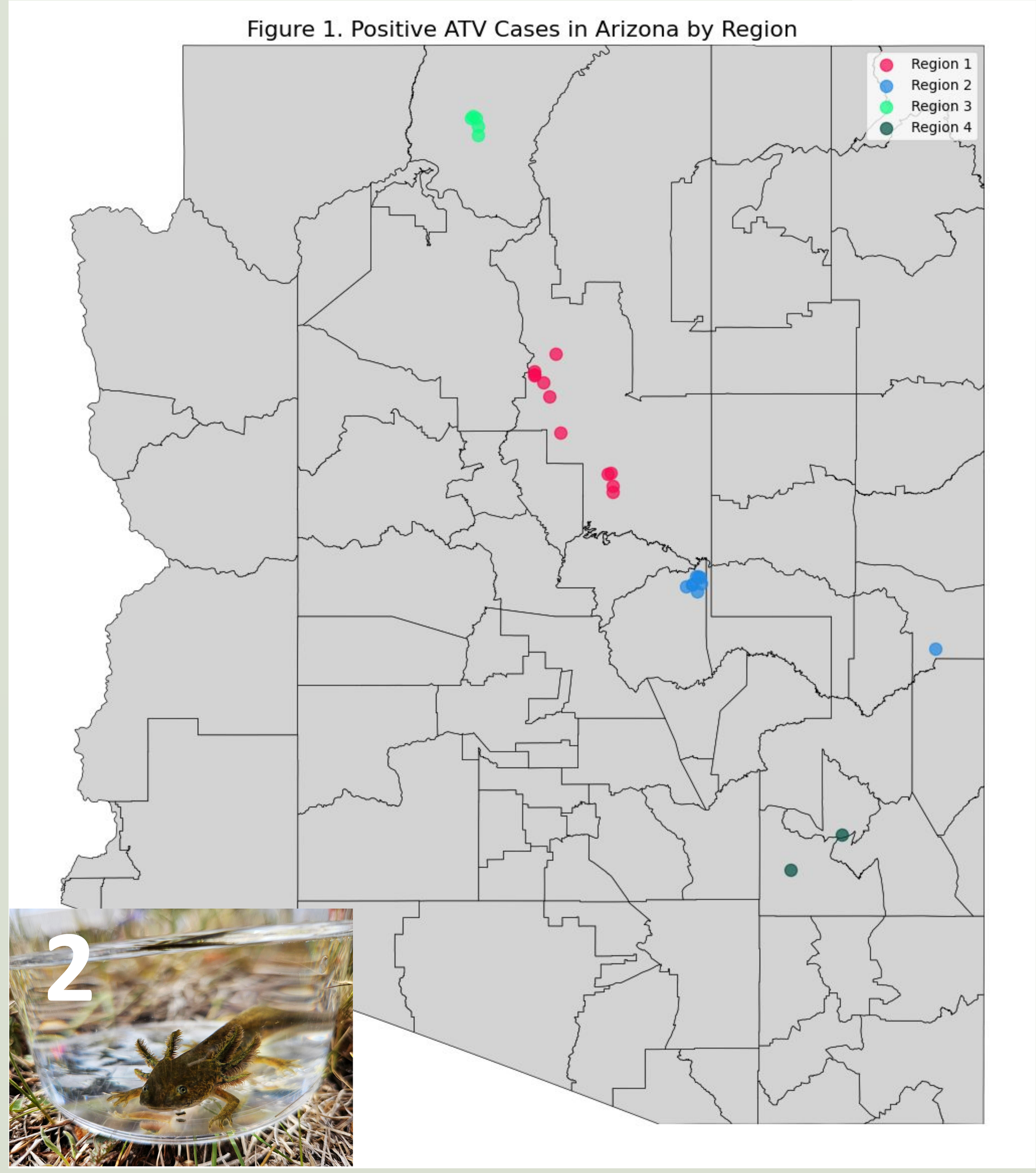
The virus Ambystoma tigrinum virus (ATV) poses a significant threat to western tiger salamanders (*Ambystoma mavortium*) in Arizona, contributing to documented amphibian population declines.

- Temperature changes may impact disease dynamics, by influencing factors such as viral replication and host immunity
- ATV spreads through direct contact between individuals and exposure to contaminated water
- Geographical or spatial variation may also influence transmission due to differences in location climate, population density, and proximity to other sites
- Asymptomatic adult carriers may create long-term disease reservoirs in the environment

This study will predict how spatial differences, such as climate, drive Ambystoma tigrinum virus (ATV) outbreaks in Arizona salamanders. **We hypothesize that temperature-driven mechanisms and landscape characteristics significantly influence infection prevalence across different regions and years.** By identifying high-risk conditions, it supports conservation strategies for protecting vulnerable amphibian populations.

OBJECTIVES

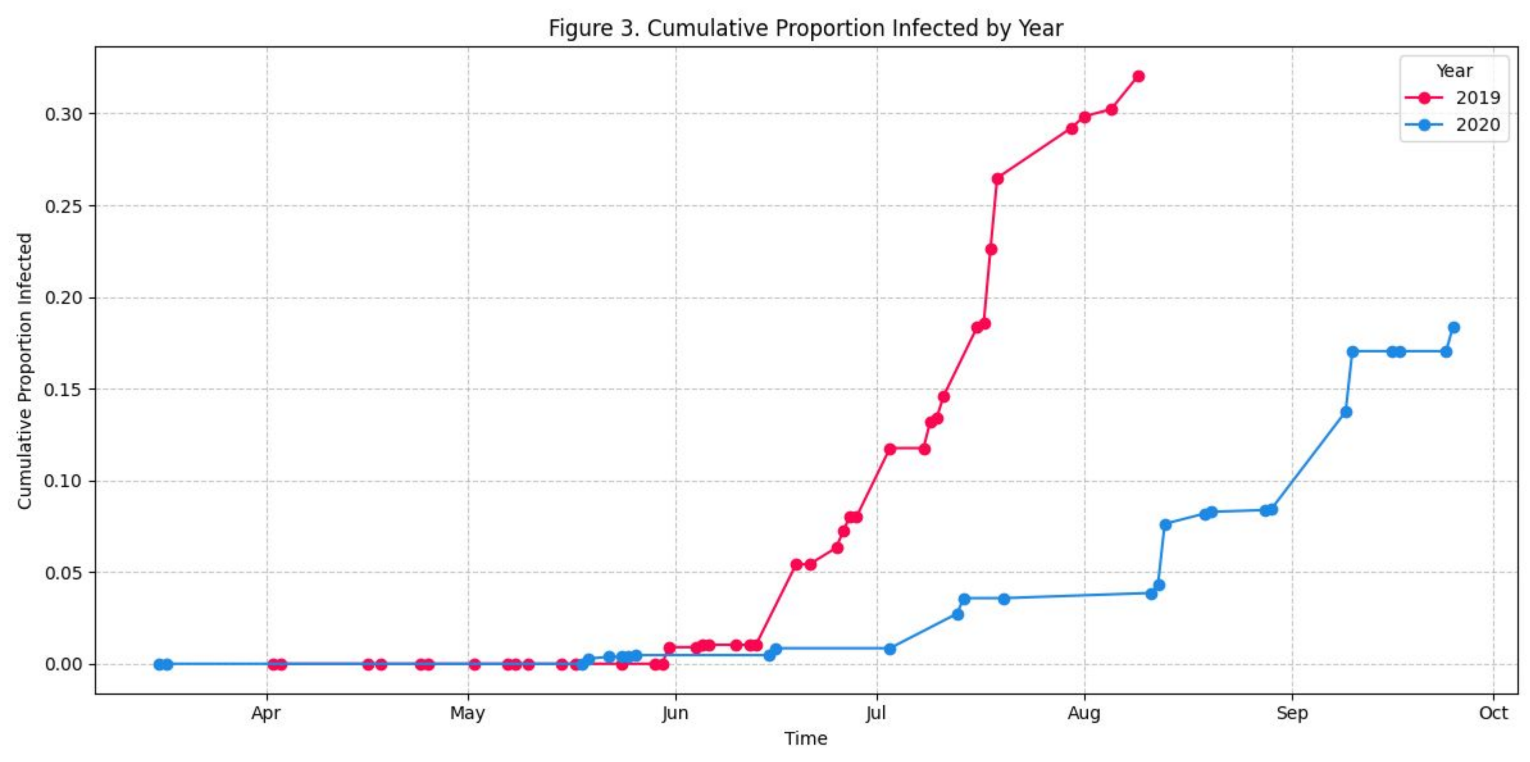
- Quantify how regional differences (climate, density, site proximity) impact disease dynamics
- Develop a model to predict the effects of temperature on ATV disease dynamics
- Investigate how temperature-dependent viral persistence influences transmission
- Validate predictions against field data from documented outbreaks

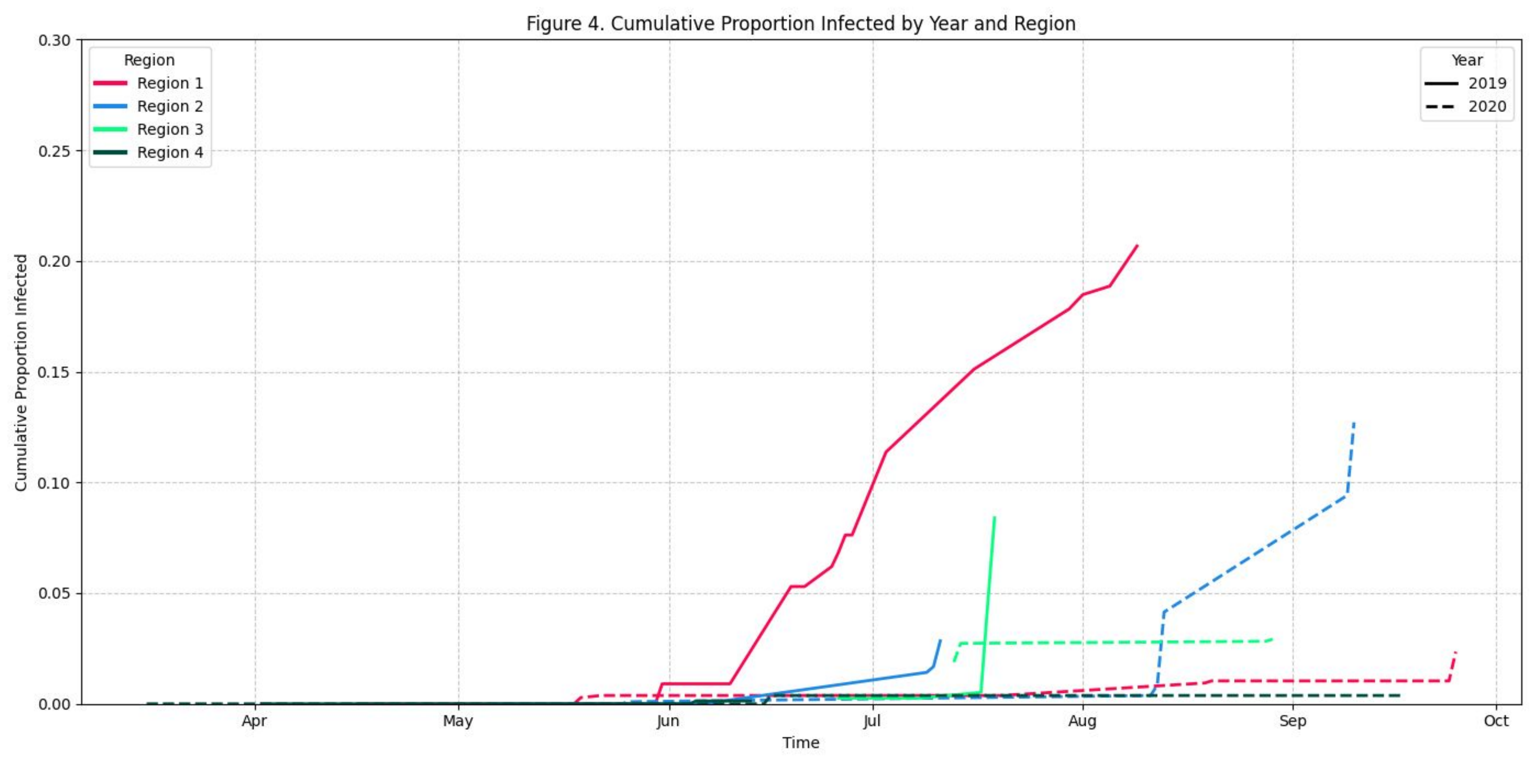


METHODS

Our study has focused on documenting and analyzing ATV outbreak patterns in Arizona salamander populations:

- Documented ATV-positive salamander sites from swabbing data in 2019 & 2020
- We have collected multi-year infection data across Arizona (Fig. 1 & 3)
- Analyzed regional differences in disease prevalence (Fig. 4)
- Identified key environmental variables associated with outbreak events including seasonal water temperature, breeding pond persistence, precipitation patterns, and elevation





RESULTS

Figure 1: Location of ATV sites across Arizona separated by 4 distinct geographical regions. Regions are defined by distance, geographical barriers, and significant differences in environmental conditions, such as average temperature.

Image 2: Juvenile aquatic western tiger salamanders in temporary sampling cups during disease sample collection in the field. Photo credit: Kelsey Banister

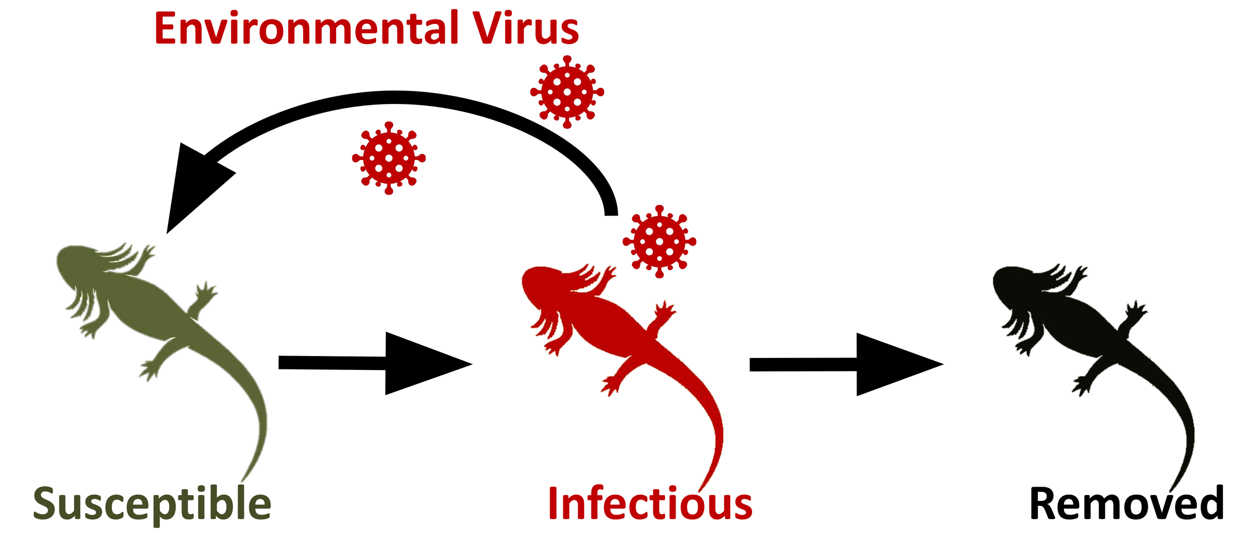
Figure 3: Depicts the cumulative proportion of hosts infected over time, by year. We observe a higher peak in the proportion of infected hosts in 2019 compared to 2020, with peak infection prevalence occurring earlier in the season in 2019, suggesting large year-to-year variation in ATV dynamics.

Figure 4: Depicts the cumulative proportion of hosts infected over time, by region and year. Northern regions (1 and 3) had a higher proportion infected than mogollon rim or southern regions (2 and 4) in 2019. However, this pattern reverses in 2020 with region 2 exhibiting the highest proportion of infected hosts. This dramatic shift in regional patterns between consecutive years highlights the complex spatial and temporal dynamics of ATV transmission.

MODEL DEVELOPMENT

Building on our current data, we will develop a comprehensive spatially-explicit Susceptible-Infected-Removed-Virus (SIRV) model using the **epymorph Python package** that integrates a multi-scaled modeling approach:

- **Individual Level:** infection dynamics within a single host (immune response vs. viral replication)
- **Population Level:** transmission dynamics between hosts within a single population
- **Spatial Level:** transmission dynamics within and across regions, such as movement between populations

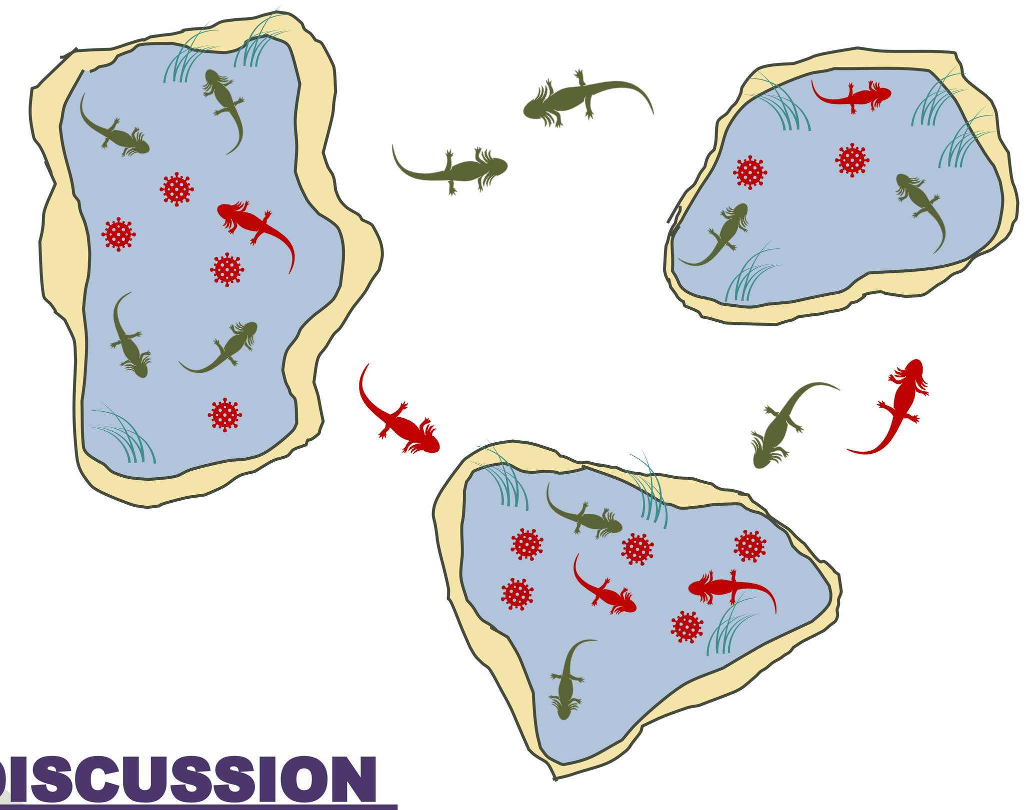


The SIRV model categorizes salamanders into three categories: **susceptible to infection (S)**, **infectious (I)**, and **removed through recovery or mortality (R)**. Where infectious individuals **shed virus from their skin back into the environment (V)**. We track how the proportion of salamanders in each category changes over time while accounting for the following key factors:

- Temperature-dependant transmission rates may vary within and across regions
- Environmental viral persistence in water and sediment
- Chronically infected adults as disease reservoirs
- Variation in seasonal host movement
- Site proximity and connectivity influencing host movement patterns

Model validation and applications:

- Calibration against field data from documented outbreaks
- Simulating multiple climate and landscape fragmentation scenarios
- Sensitivity analysis to identify key drivers of epizootic outcomes
- Projection of high-risk conditions to inform conservation strategies



DISCUSSION

Our preliminary results show regional and temporal variation in ATV infections across Arizona salamander populations. Year-to-year fluctuations likely reflect climate-driven mechanisms similar to those documented in other disease systems, where temperature influences both viral replication and host immunity (Altizer et al., 2006). Notably, regions 1 and 2 show the highest proportion of infected individuals, which is likely due to their intermediate temperatures (18-22°C) – conditions that are close to the optimal range for ATV replication. In contrast, the northernmost and southernmost regions, which tend to have more extreme temperatures, fall outside this optimal temperature range and show lower infection prevalence.

These spatial differences in temperature likely influence the severity and spread of epizootics, as temperature affects both the virus's ability to replicate and the host's immune responses (Rojas et al. 2005). Additionally, other spatial factors such as geographic barriers and habitat connectivity combine with regional climate to further contribute to the observed differences. Future research will incorporate host density, environmental viral persistence, and adult movement patterns to develop a model that captures ATV transmission under climate change.

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